CSCI 49376 Assignment 1 Report

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Part 1. Select and Examine the Original Dataset

1. Dataset selected: Diabetes 130-US Hospitals for Years 1999-2008 (Kaggle: brandao/diabetes)
2. CSV file used: `diabetic_data.csv`
3. Total rows: 101,766
4. Total columns: 50 (24 ignored, 26 used)
5. Attributes:
    1. encounter_id
    2. patient_nbr
    3. race
    4. gender
    5. age
    6. weight
    7. admission_type_id
    8. discharge_disposition_id
    9. admission_source_id
    10. time_in_hospital
    11. payer_code
    12. medical_specialty
    13. num_lab_procedures
    14. num_procedures
    15. num_medications
    16. number_outpatient
    17. number_emergency
    18. number_inpatient
    19. diag_1
    20. diag_2
    21. diag_3
    22. number_diagnoses
    23. max_glu_serum
    24. A1Cresult
    25. diabetesMed
    26. readmitted

Ignored (24): metformin, repaglinide, nateglinide, chlorpropamide, glimepiride, acetohexamide, glipizide, glyburide, tolbutamide, pioglitazone, rosiglitazone, acarbose, miglitol, troglitazone, tolazamide, examide, citoglipton, insulin, glyburide-metformin, glipizide-metformin, glimepiride-pioglitazone, metformin-rosiglitazone, metformin-pioglitazone, change. 6. Brief description of what each attribute represents: 1. encounter_id: Unique ID of one hospital stay 2. patient_nbr: ID of a patient (a patient can have several stays) 3. race: Caucasian, AfricanAmerican, Hispanic, Asian, Other 4. gender: Female, Male, Unknown/Invalid 5. age: Age group 6. weight: Weight band in pounds 7. admission_type_id: Code for the type of admission (e.g. emergency, elective) 8. discharge_disposition_id: Code for where the patient went at discharge (e.g. home) 9. admission_source_id: Code for where the patient came from (e.g. referral, ER) 10. time_in_hospital: Days in the hospital (1–14) 11.

payer_code: Insurance code 12. medical_specialty: Specialty of the admitting doctor 13. num_lab_procedures: Lab tests during the stay 14. num_procedures: Other procedures during the stay 15. num_medications: Distinct medications given 16. number_outpatient: Outpatient visits in the prior year 17. number_emergency: Emergency visits in the prior year 18. number_inpatient: Inpatient visits in the prior year 19. diag_1: Primary diagnosis (ICD-9 code) 20. diag_2: Secondary diagnosis (ICD-9 code) 21. diag_3: Additional diagnosis (ICD-9 code) 22. number_diagnoses: Number of diagnoses recorded 23. max_glu_serum: Glucose test: >200, >300, Norm, None (not measured) 24. A1Cresult: HbA1c test: >7, >8, Norm, None (not measured) 25. diabetesMed: Any diabetes medication prescribed: Yes/No 26. readmitted: NO, >30 (after 30 days), <30 (within 30 days) 7. Possible identifiers or keys: 1. encounter_id: key of a stay. One value per row (101,766). 2. patient_nbr: ID of a patient (71,518 patients). A patient can have several stays. 3. admission_type_id, discharge_disposition_id, admission_source_id: codes (8, 26, and 17 values). 4. diag_1, diag_2, diag_3: diagnosis codes. Many stays share a code. 8. Missing, unknown or incomplete values: 1. weight (?, 96.9%) 2. max_glu_serum: None (not measured, 94.7%) 3. A1Cresult: None (not measured, 83.3%) 4. medical_specialty (?, 49.1%) 5. payer_code (?, 39.6%) 6. race (?, 2,273 rows, 2.2%) 7. gender: Unknown/Invalid (3) 8. diag_1 (?, 21), diag_2 (?, 358), diag_3 (?, 1,423)

```
All of these are stored as NULL in the database.
```

Inconsistencies: 123 patients have two different race values and 3 have two genders across stays

Part 2. Normalize the Dataset

In short: 1NF puts each diagnosis on its own row. 2NF moves the stay columns to ENCOUNTER, so each stay is stored once.

Example used throughout: stay 149190 (patient 55629189). **Stay columns** = the columns that describe one hospital visit: race, gender, age, weight, admission_type_id, discharge_disposition_id, admission_source_id, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted. Race and gender are stay columns. They can differ across a patient's stays.

Original relation

1. Structure: DIABETIC_DATA, one table, one row per stay (101,766 rows). diag_1, diag_2 and diag_3 are on that row.
2. Key: encounter_id.
3. Functional dependencies: encounter_id → every other column.
4. Problem: diag_1, diag_2 and diag_3 are a repeating group. Three columns, same kind of value.
5. Change: one row per diagnosis. Add diag_position, 1, 2 or 3.
6. Form: not 1NF, because of the repeating group.

encounter_id	patient_nbr	race	gender	age	time_in_hospital	diag_1	diag_2	diag_3	readmitted
149190	55629189	Caucasian	Female	[10-20)	3	276	250.01	255	>30

1NF relation

1. Structure: one row per diagnosis. diag_1, diag_2 and diag_3 become icd9_code, plus diag_position. Stay columns still repeat on each row.
2. Key: (encounter_id, diag_position). 5,492 stays list the same code twice, so icd9_code is not in the key.
3. Functional dependencies: (encounter_id, diag_position) → icd9_code. encounter_id → the stay columns.

4. Problem: partial dependency. Stay columns depend on encounter_id only, so they repeat on every diagnosis row.
5. Change: move the stay columns to ENCOUNTER. Diagnoses stay in ENCOUNTER_DIAGNOSIS.
6. Form: 1NF. Every cell holds one value.

encounter_id	diag_position	icd9_code	patient_nbr	race	time_in_hospital	readmitted
149190	1	276	55629189	Caucasian	3	>30
149190	2	250.01	55629189	Caucasian	3	>30
149190	3	255	55629189	Caucasian	3	>30

2NF relations

1. Structure: ENCOUNTER(encounter_id, patient_nbr, stay columns) and ENCOUNTER_DIAGNOSIS(encounter_id, diag_position, icd9_code).
2. Keys: ENCOUNTER: encounter_id. ENCOUNTER_DIAGNOSIS: (encounter_id, diag_position). encounter_id points to ENCOUNTER.
3. Functional dependencies: encounter_id → the stay columns, in ENCOUNTER. (encounter_id, diag_position) → icd9_code, in ENCOUNTER_DIAGNOSIS.
4. Problem: none.
5. Change: none.
6. Form: 2NF. Every non-key column depends on the whole key.

ENCOUNTER

encounter_id	patient_nbr	race	gender	admission_type_id	age	time_in_hospital	readmitted
149190	55629189	Caucasian	Female	1	[10-20)	3	>30

ENCOUNTER_DIAGNOSIS

encounter_id	diag_position	icd9_code
149190	1	276
149190	2	250.01
149190	3	255

Final tables

ENCOUNTER and ENCOUNTER_DIAGNOSIS are the 2NF tables. PATIENT is the patient id. DIAGNOSIS stores each code once. The three code tables store each id and its description.

Table	Attributes	Primary key	Foreign key(s)
PATIENT	patient_nbr	patient_nbr	—
ENCOUNTER	encounter_id, patient_nbr, race, gender, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	encounter_id	patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id

Table	Attributes	Primary key	Foreign key(s)
ENCOUNTER_DIAGNOSIS	encounter_id, diag_position, icd9_code	(encounter_id, diag_position)	encounter_id, icd9_code
DIAGNOSIS	icd9_code	icd9_code	—
ADMISSION_TYPE	admission_type_id, description	admission_type_id	—
DISCHARGE_DISPOSITION	discharge_disposition_id, description	discharge_disposition_id	—
ADMISSION_SOURCE	admission_source_id, description	admission_source_id	—

Part 3. Construct an ER Diagram



Relationship	Between	Type	Why	Participation
has	PATIENT – ENCOUNTER	1:N	A patient has many stays; a stay has one patient.	Total on both sides
classifies	ADMISSION_TYPE – ENCOUNTER	1:N	A type covers many stays; a stay has one type.	Total on both sides

Relationship	Between	Type	Why	Participation
ends with	DISCHARGE_DISPOSITION – ENCOUNTER	1:N	A disposition covers many stays; a stay has one disposition.	Total on both sides
admitted from	ADMISSION_SOURCE – ENCOUNTER	1:N	A source covers many stays; a stay has one source.	Total on both sides
diagnosed with	ENCOUNTER – DIAGNOSIS	M:N	A stay has up to 3 diagnoses; a diagnosis appears in many stays. Drawn as two 1:N relationships through ENCOUNTER_DIAGNOSIS, which also stores diag_position.	Partial for ENCOUNTER (some stays have no diagnosis), total for DIAGNOSIS

Part 4. Convert the ER Diagram into Relational Tables

Relation	Attributes	Primary key	Foreign key(s)
PATIENT	patient_nbr	patient_nbr	—
ADMISSION_TYPE	admission_type_id, description	admission_type_id	—
DISCHARGE_DISPOSITION	discharge_disposition_id, description	discharge_disposition_id	—
ADMISSION_SOURCE	admission_source_id, description	admission_source_id	—
DIAGNOSIS	icd9_code	icd9_code	—
ENCOUNTER	encounter_id, patient_nbr, race, gender, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	encounter_id	patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id
ENCOUNTER_DIAGNOSIS	encounter_id, diag_position, icd9_code	(encounter_id, diag_position)	encounter_id, icd9_code

How the ER relationships are represented

ER relationship	Type	Represented as
PATIENT has ENCOUNTER	1:N	Foreign key ENCOUNTER.patient_nbr → PATIENT (on the N side, NOT NULL)
ADMISSION_TYPE classifies ENCOUNTER	1:N	Foreign key ENCOUNTER.admission_type_id → ADMISSION_TYPE (on the N side, NOT NULL)
DISCHARGE_DISPOSITION ends with ENCOUNTER	1:N	Foreign key ENCOUNTER.discharge_disposition_id → DISCHARGE_DISPOSITION (on the N side, NOT NULL)
ADMISSION_SOURCE admitted from ENCOUNTER	1:N	Foreign key ENCOUNTER.admission_source_id → ADMISSION_SOURCE (on the N side, NOT NULL)
ENCOUNTER diagnosed with DIAGNOSIS	M:N	Its own table, ENCOUNTER_DIAGNOSIS: a foreign key to each side plus diag_position

Table creation order: PATIENT, ADMISSION_TYPE, DISCHARGE_DISPOSITION, ADMISSION_SOURCE, DIAGNOSIS → ENCOUNTER → ENCOUNTER_DIAGNOSIS.

Part 5. Implement the Database Using SQL

Jas. MySQL script: `work/sql/create_tables.sql`.

```
-- Assignment 1 Part 5. MySQL.
-- Generated by work/factcheck.ipynb from diabetic_data.csv.
-- Parents are created before the tables that reference them.

DROP DATABASE IF EXISTS hospital;
CREATE DATABASE hospital;
USE hospital;

CREATE TABLE patient (
    patient_nbr INT PRIMARY KEY
);

CREATE TABLE admission_type (
    admission_type_id INT PRIMARY KEY,
    description VARCHAR(160)
);

CREATE TABLE discharge_disposition (
    discharge_disposition_id INT PRIMARY KEY,
    description VARCHAR(160)
);

CREATE TABLE admission_source (
    admission_source_id INT PRIMARY KEY,
    description VARCHAR(160)
);

CREATE TABLE diagnosis (
    icd9_code VARCHAR(10) PRIMARY KEY
);

CREATE TABLE encounter (
    encounter_id INT PRIMARY KEY,
    patient_nbr INT NOT NULL,
    race VARCHAR(20),
    gender VARCHAR(20),
    admission_type_id INT NOT NULL,
    discharge_disposition_id INT NOT NULL,
    admission_source_id INT NOT NULL,
    age VARCHAR(10) NOT NULL,
    weight VARCHAR(10),
    payer_code VARCHAR(10),
    medical_specialty VARCHAR(40),
    time_in_hospital INT NOT NULL CHECK (time_in_hospital BETWEEN 1 AND 14),
    num_lab_procedures INT NOT NULL,
    num_procedures INT NOT NULL,
    num_medications INT NOT NULL,
    number_outpatient INT NOT NULL,
    number_emergency INT NOT NULL,
    number_inpatient INT NOT NULL,
    number_diagnoses INT NOT NULL,
    max_glu_serum VARCHAR(10),
    A1Cresult VARCHAR(10),
    diabetesMed VARCHAR(10) NOT NULL,
```

```
    readmitted VARCHAR(10) NOT NULL,
    CONSTRAINT fk_enc_patient_nbr FOREIGN KEY (patient_nbr) REFERENCES patient (patient_nbr),
    CONSTRAINT fk_enc_admission_type_id FOREIGN KEY (admission_type_id) REFERENCES
admission_type (admission_type_id),
    CONSTRAINT fk_enc_discharge_disposition_id FOREIGN KEY (discharge_disposition_id)
REFERENCES discharge_disposition (discharge_disposition_id),
    CONSTRAINT fk_enc_admission_source_id FOREIGN KEY (admission_source_id) REFERENCES
admission_source (admission_source_id)
);

CREATE TABLE encounter_diagnosis (
    encounter_id INT NOT NULL,
    diag_position INT NOT NULL CHECK (diag_position IN (1, 2, 3)),
    icd9_code VARCHAR(10) NOT NULL,
    PRIMARY KEY (encounter_id, diag_position),
    FOREIGN KEY (encounter_id) REFERENCES encounter (encounter_id),
    FOREIGN KEY (icd9_code) REFERENCES diagnosis (icd9_code)
);
```

Part 6. Import the CSV Data

Jas. MySQL script: `work/sql/import_data.sql`. The path is CSV → staging table → the seven normalized tables. No CSV row is typed by hand.

1. **Staging table.** `CREATE TABLE staging` with one column per CSV column.
2. **Load the CSV.** `LOAD DATA LOCAL INFILE 'diabetic_data.csv' INTO TABLE staging ... IGNORE 1 ROWS` reads all 101,766 rows.
3. **Code tables first.** ADMISSION_TYPE, DISCHARGE_DISPOSITION and ADMISSION_SOURCE get each code that appears in the CSV, with its description from the published IDs_mapping labels.
4. **PATIENT.** `INSERT INTO patient SELECT DISTINCT patient_nbr FROM staging`: one row per patient.
5. **DIAGNOSIS.** The distinct codes from diag_1, diag_2 and diag_3 combined with `UNION`, skipping `?`.
6. **ENCOUNTER.** One row per staging row. `?` and Unknown/Invalid become NULL, and `None` on the two lab columns becomes NULL.
7. **ENCOUNTER_DIAGNOSIS.** Three `SELECT`s joined with `UNION ALL`: diag_1 as position 1, diag_2 as position 2, diag_3 as position 3, skipping `?`. This is the 1NF step done in SQL.

Parents are filled before children, so every foreign key has a matching row.

Part 7. Verify the Database Using SQL

Jas. MySQL script: `work/sql/verification_queries.sql`. These are the results of the same statements run against the loaded database.

1. Tables contain the expected rows.
    1. patient: 71,518
    2. encounter: 101,766
    3. encounter_diagnosis: 303,496
    4. diagnosis: 915
    5. admission_type: 8
    6. discharge_disposition: 26
    7. admission_source: 17
2. Primary keys do not repeat. Each duplicate query returns no rows.

3. Foreign keys point at a real parent. Each orphan query returns 0.
4. The join rebuilds a stay. Encounter 2278392, the first row of the CSV, comes back as Caucasian, Female, age [0-10), 1 day, diagnosis position 1 = 250.83, readmitted NO. diag_2 and diag_3 were ?, so they have no diagnosis rows.
5. The ENCOUNTER columns match the CSV on every row. Race and gender are on the stay. ? and Unknown/Invalid are NULL.